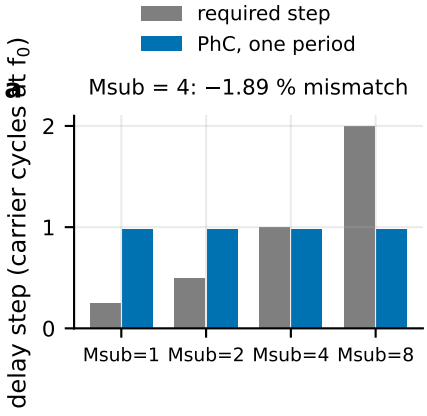
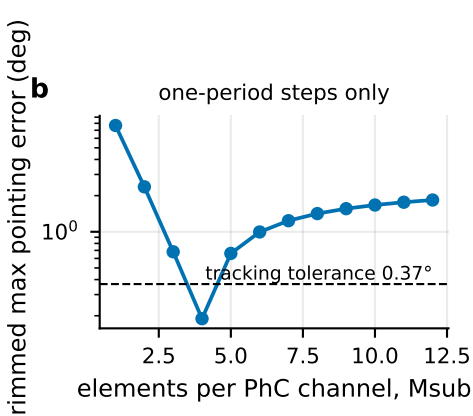
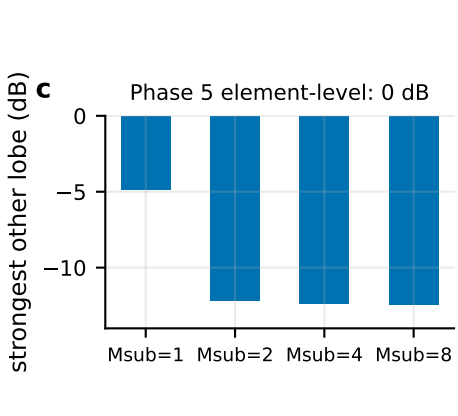

a
required step
PhC, one period
Msub = 4: −1.89 % mismatch
delay step (carrier cycles at f_0)
2
1
0
Msub=1
Msub=2
Msub=4
Msub=8


b
one-period steps only
trimmed max pointing error (deg)
10^0
tracking tolerance 0.37°
2.5
5.0
7.5
10.0
12.5
elements per PhC channel, Msub


c
Phase 5 element-level: 0 dB
strongest other lobe (dB)
0
−5
−10
Msub=1
Msub=2
Msub=4
Msub=8